① 推导 抽象 独立思考

② 以例为主 边做边学

③ pd 范式 eg. 例子

阶数分析、微元法 用处：偏慢

- 代数
- 微分
- 特情

一、阶数分析

1. 代数方程

pd. $\boxed{f(x) = g(x)}$

$f^0 + f'x + f^2x^2 \cdots = g^0 + g'x + \cdots$

$(f^0 - g^0) + (f' - g')x + (f^2 - g^2)x^2 = 0$

$f^0 = g^0 \quad f' = g' \quad f^2 = g^2 \cdots$

$f(x) - g(x) \qquad F(x) = 0$

$f(x) = f_0 + f_1 + f_2 + \cdots$

$= f^0 + f'x + f^2x^2 + \cdots$

eg1. $\alpha x^3 + x^2 - 1 = 0 \quad (\alpha << 1)$ 求 $x(\alpha)$ 正根（保至 α^2 项）

$x = x^0 + x'\alpha + x^2\alpha^2$

$\alpha(x^0 + x'\alpha + x^2\alpha^2)^3 + (\quad)^2 - 1 = 0$

$(\quad) + (\quad)\alpha + (\quad)\alpha^2 + O(\alpha^3) = 0$

pd2. “微扰法”：一阶一阶解.

n阶 $\leq$ n-1阶

$\alpha x^3 + x^2 - 1 = 0$

$x^0 = 1$

$x = x_0 + x_1 = 1 + x'\alpha$

$\alpha(1 + x'\alpha)^3 + (1 + x'\alpha)^2 - 1 = 0$

$(\quad)\alpha = 0$

$\alpha + 1 + 2x'\alpha - 1 = 0$

$(1 + 2x')\alpha = 0$

$x' = -\frac{1}{2}$

$$X = 1 - \frac{1}{2}\alpha + X^2\alpha^2$$

$$\alpha\left(1 - \frac{1}{2}\alpha + X^2\alpha^2\right)^3 + \left(1 - \frac{1}{2}\alpha + X^2\alpha^2\right)^2 - 1 = 0$$

$$\alpha\left(1 - \frac{3}{2}\alpha\right) + 1 + 2\left(-\frac{1}{2}\alpha + X^2\alpha^2\right) + \left(-\frac{1}{2}\alpha + X^2\alpha^2\right)^2 - 1 = 0$$

$$-\frac{3}{2}\alpha^2 + 2X^2\alpha^2 + \frac{1}{4}\alpha^2 = 0$$

$$X^2 = \frac{5}{8}$$

$$\chi = 1 - \frac{1}{2}\alpha + \frac{5}{8}\alpha^2$$

2. 微分方程

eg1 $\ddot{x} + \omega^2 x + \alpha x^2 = 0$. $(\alpha \ll 1 \quad \omega \sim 1)$.

$x = x_0 + O(1)$.

$$\ddot{x} + \omega^2 x = 0.$$

$x_0 = A\cos(\omega t + \varphi)$.

$x = A\cos(\omega t + \varphi) + x_1 + O_2$

$\ddot{x}_1 + \omega^2 x_1 + \alpha A\cos(\omega t + \varphi) = 0$

3. 特殊技巧

3.1 $\underset{\triangle}{f(y)} = \underset{\triangle}{g(x)}$. $y(x)$ 显式？

$$f^1 y + f^2 y^2 + f^3 y^3 \cdots = g^1 x + g^2 x^2 + g^3 x^3 \cdots$$

$y_{(x)} = \alpha_1 X + \alpha_2 X^2 + \alpha_3 X^3 \cdots$

* $f^1 y = g^1 x + g^2 x^2 + g^3 x^3 \cdots - (f^2 y^2 + f^3 y^3 \cdots)$

$y_{(x)} = \alpha_1 X + O(x^2)$.

$(\alpha_1 X + \alpha_2 X^2 + \cdots)^2$

$f^1 y = g^1 x$

$y_1 = \frac{g^1}{f^1} x$

$$y_{(x)} = \frac{g^1}{f^1} x + \alpha_2 x^2 + O(x^3).$$

$$f^1 y = g^1 x + g^2 x^2 - (f^2 \alpha_1^2 x^2).$$

$$y = \frac{g^1}{f^1} x + \frac{g^2 - f^2\alpha_1^2}{f^1} x^2$$

$$y = y_1 + y_2 + \alpha_3 x^3 + O(x^4)$$

eg. $n_1 \sin\theta = n_2 \sin\varphi$ 求 $\theta(\varphi)$ 保 φ^3.

$$n_1(\theta - \frac{\theta^3}{6}) = n_2(\varphi - \frac{\varphi^3}{6})$$

$$n_1\theta = n_2\varphi - \frac{n_2}{6}\varphi^3 + \frac{n_1}{6}\theta^3$$

$$\theta_1 = \frac{n_2}{n_1}\varphi$$

$$\theta = \frac{n_2}{n_1}\varphi + \theta_3$$

$$n_1\theta_3 = -\frac{n_2}{6}\varphi^3 + \frac{n_1}{6}(\frac{n_2}{n_1})^3\varphi^3$$

$$\theta_3 = (-\frac{n_2}{6n_1} + \frac{n_2^3}{6n_1^3})\varphi^3$$

$$\theta = \frac{n_2}{n_1}\varphi + (\frac{n_2^3}{6n_1^3} - \frac{n_2}{6n_1})\varphi^3$$

3.2 有一阶大的情况

$\delta \to 0$ $n\delta \sim 1$

$$(1+\delta)^n = 1 + n\delta + \frac{1}{2}n(n-1)\delta^2 + \frac{1}{6}n(n-1)(n-2)\delta^3 + \cdots$$

3.3 积分

pd. $f(y)dy = f(x)dx$

$$(A_0 + A_1 y + A_2 y^2)dy = (B_0 + B_1 x + B_2 x^2)dx$$

$$A_0 y + \frac{1}{2}A_1 y^2 + \frac{1}{3}A_2 y^3 \cdots = B_0 x + \frac{1}{2}B_1 x^2 + \frac{1}{3}B_2 x^3 \cdots$$

$y(x)$ 到 x^3.

eg $\frac{d\theta}{\cos\theta} = k\,dx$ $\theta(x)$

$$\frac{d\theta}{\cos\theta_0} = k\,dx$$

$$\Delta\theta = \cos\theta_0 k\,\Delta x$$

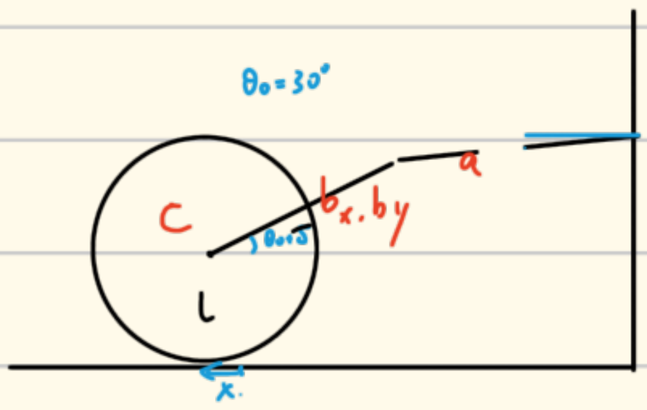
3.4

$$\sqrt{mc^2 - \frac{1}{2}k(R^2-r^2)} \times \sqrt{mc^2\left(1 - \frac{k(R^2-r^2)}{2mc^2}\right)}$$

二、微元法

目标：消约束，找牵联关系。

e.g.



m

X. δ. α

$\delta(\alpha)$ $X(\alpha)$.

b_x, b_y, α, X.

T. V.

$$T_a = \frac{1}{2}\cdot\frac{1}{3}\cdot 4ml^2\dot{\alpha}^2$$

$$T_b = \frac{1}{2}m(\dot{b_x}^2 + \dot{b_y}^2) + \frac{1}{2}\cdot\frac{1}{3}\cdot 4ml^2\cdot\dot{\delta}^2$$

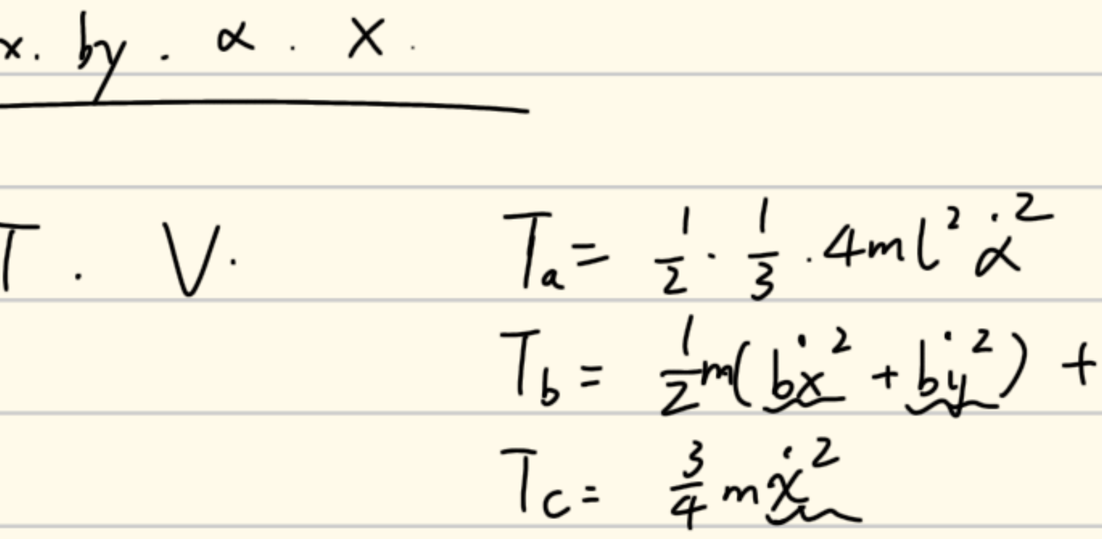
$$T_c = \frac{3}{4}m\dot{x}^2$$

$V_a = \cdots$

$V_b ----$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\alpha}}\right) - \frac{\partial L}{\partial \alpha} = 0 \quad \cdots \quad \frac{\partial T_a}{\partial \dot{\alpha}}\bigg|_{\dot{\alpha}=0} \quad \frac{\partial V_a}{\partial \alpha}\bigg|_{\alpha=0}$$

$$X + (\sqrt{3}+2)L = 2L\cos(\theta_0+\delta) + 2L\cos\alpha \qquad ①$$



$$2L\sin(\theta_0+\delta) + 2L\sin\alpha = L \qquad ②$$

pdl. $\boxed{X} = \boxed{X'\alpha} + X^2\alpha^2 + X^3\alpha^3$

$\dot{X} = X'\dot{\alpha} + 2X^2\alpha\dot{\alpha} + 3X^3\alpha^2\dot{\alpha}$

$\ddot{X} = \ldots\ldots$

$\frac{\partial}{\partial\dot{\alpha}}\dot{\alpha}|_{t=0} = 0$

$\alpha = 0 \quad \dot{\alpha} = 0$

$X = X'\alpha$

$\delta = \delta'\alpha$

$\sin(\theta_0+\delta) + \sin\alpha = \frac{1}{2}$

$\cos\theta_0\,\delta + \alpha = 0$

$\delta = -\frac{1}{\cos\theta_0}\alpha$

$\delta = -\frac{2\sqrt{3}}{3}\alpha$ ①

$X + (\sqrt{3}+2)L = 2L\cos(\theta_0+\delta) + 2L\cos\alpha$ ①

$X = -2L\sin\theta_0\,\delta$

$X = \frac{2\sqrt{3}}{3}L\alpha$ ②

$X = X'\alpha + X^2\alpha^2 + X^3\alpha^3$

$\dot{X} = X'\alpha + 2X^2\dot{\alpha}\alpha + 3X^3\alpha^2\dot{\alpha}$

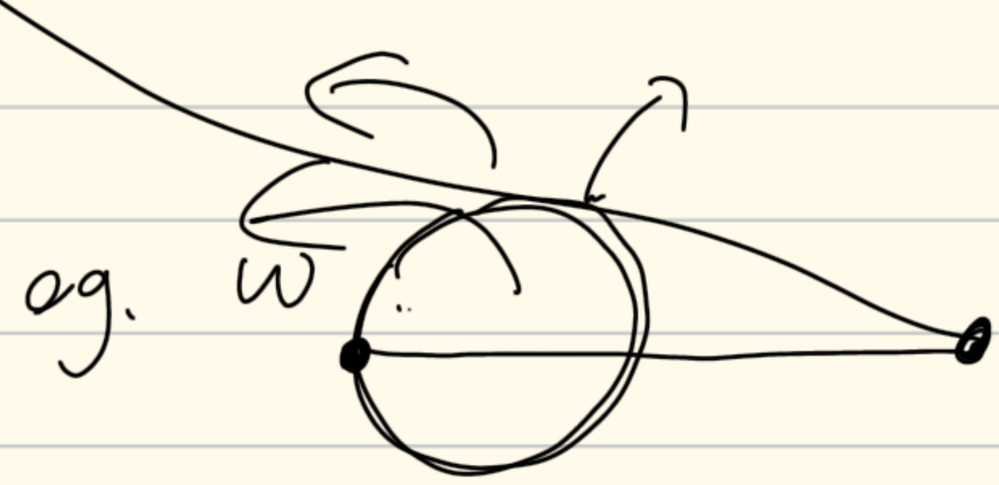
$\dot{\alpha} \neq 0$

$\frac{\partial L}{\partial\alpha} = 2X^2\dot{\alpha}$

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\alpha}}\right) = 2X^2\ddot{\alpha}$

$\frac{\partial U}{\partial\alpha} = 2X^2\alpha$ $\alpha=0$

eg.



$\chi(q_1 \cdot q_2 \cdot q_3 \cdots\cdots)$.

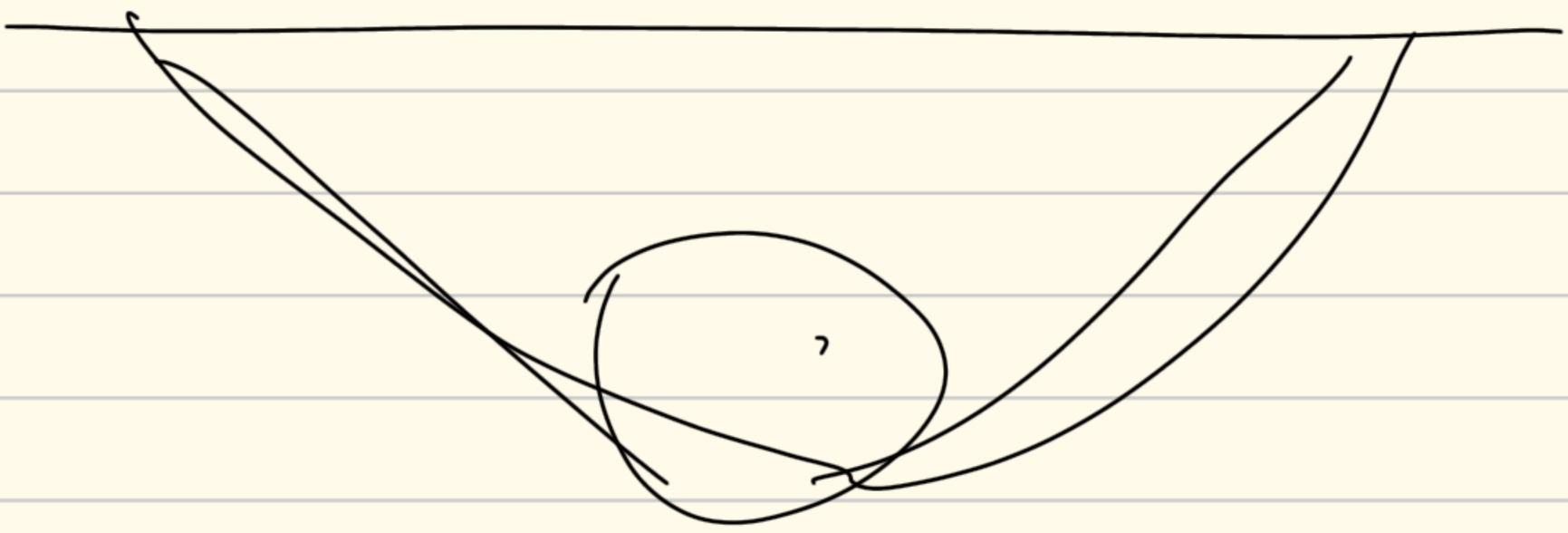
$$\boxed{\Delta q_i \cdot \frac{\partial}{\partial q_i} \equiv \sum_i \Delta q_i \cdot \frac{\partial}{\partial q_i}}$$

$$X = X_0 + (\Delta q_i \cdot \frac{\partial}{\partial q_i})\chi + \frac{1}{2!}(\Delta q_i \cdot \frac{\partial}{\partial q_i})^2 \chi^2 + \frac{1}{3!}(\Delta q_i \cdot \frac{\partial}{\partial q_i})^3 \chi^3 \cdots\cdots$$

$$X = X_0 + \alpha_i \Delta q_i + \beta_{ij} \Delta q_i \Delta q_j + \gamma_{ijk} \Delta q_i \Delta q_j \Delta q_k \cdots\cdots$$

$$\dot{x} = \dot{x}_0 + \alpha_i \Delta \dot{q}_i + \beta_{ij} \Delta q_i \Delta \dot{q}_j + \beta_{ij} \Delta \dot{q}_i \Delta \dot{q}_j + \cdots\cdots$$

All i : $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$ | $q_i = 0$



更正（明日将提及）

有个地方讲错了



$$x = x'\alpha + x^2\alpha^2 + x^3\alpha^3 + O_{(\alpha^4)}$$

$$\dot{x} = x'\dot{\alpha} + 2x^2\alpha\dot{\alpha} + 3x^3\alpha^2\dot{\alpha} + O_{(\alpha^4)}$$

$$\ddot{x} = x'\ddot{\alpha} + 2x^2\dot{\alpha}^2 + 2x^2\alpha\ddot{\alpha} + 6x^3\alpha\dot{\alpha}^2 + 3x^3\alpha^2\ddot{\alpha} + O_{(\alpha^4)}$$

牛顿视角：

牛顿方程：$\ddot{x}, x, \ddot{\alpha}, \alpha, (\dot{x}, \dot{\alpha})$

质心牛二

角动量

$t=0$ 时

若 $t=0$ 时 $\alpha=0$，$\dot{\alpha}=0$，$\ddot{\alpha}\neq 0$

则 $x|_{t=0} = 0$

$\dot{x}|_{t=0} = 0$

$\ddot{x}|_{t=0} = x'\ddot{\alpha}$　　保 $x=x'\alpha$ 即可

若 $t=0$ 时，$\alpha=0$，$\dot{\alpha}\neq 0$，$\ddot{\alpha}\neq 0$

则 $x|_{t=0} = 0$

$\dot{x}|_{t=0} = x'\dot{\alpha}$

$\ddot{x}|_{t=0} = x'\ddot{\alpha} + 2x^2\dot{\alpha}^2$　　保 $x = x'\alpha + x^2\alpha^2$ 即可

拉格朗日视角：（比牛顿视角更烦）　　没有那么显然！

$L(\dot{\alpha}, \alpha)$　　~~$L(\dot{\alpha}, \alpha, \dot{x}, x)$~~

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\alpha}}\right) - \frac{\partial L}{\partial \alpha} = 0$$　后还会剩下的有什么？

$$\frac{\partial L}{\partial \dot{\alpha}} = \left(\frac{\partial L}{\partial \dot{x}}\right)_{\alpha}\frac{\partial \dot{x}}{\partial \dot{\alpha}} + \left(\frac{\partial L}{\partial \dot{\alpha}}\right)_{x}$$

$$\frac{\partial L}{\partial \alpha} = \left(\frac{\partial L}{\partial x}\right)_{\alpha}\frac{\partial x}{\partial \alpha} + \left(\frac{\partial L}{\partial \alpha}\right)_{x}$$

$$= \left(\frac{\partial L}{\partial \dot{x}}\right)_{\alpha}\cdot\left(x' + 2x^2\alpha + 3x^3\alpha^2 \cdots\right) + \frac{\partial L}{\partial \dot{\alpha}}$$

.....

真没那么显然 QwQ　　还是用牛顿视角吧